

Foundations of Naive Bayes classifier

[Naive Bayes classifier]

$$P(C_k | x) \propto P(C_k) \prod_{i=1}^n P(x_i | C_k)$$

1.0 Content Block

The left-hand side of this equation is the log-odds, or logit, the quantity predicted by the linear model that underlies logistic regression. Since naive Bayes is also a linear model for the two "discrete" event models, it can be reparametrised as a linear function $b + w^T x > 0$. Obtaining the probabilities is then a matter of applying the logistic function to $b + w^T x$, or in the multiclass case, the softmax function. Discriminative classifiers have lower asymptotic error than generative ones; however, research by Ng and Jordan has shown that in some practical cases naive Bayes can outperform logistic regression because it reaches its asymptotic error faster.

2.0 Content Block

$\Pr(S | W)$ is the spamicity of this word. (Demonstration:) This corrected probability is used instead of the spamicity in the combining formula. This formula can be extended to the case where n is equal to zero (and where the spamicity is not defined), and evaluates in this case to $\Pr(S)$.

3.0 Content Block

$\ln \frac{p(S | D)}{p(\neg S | D)} = \ln \frac{p(S)}{p(\neg S)} + \sum_i \ln \frac{p(w_i | S)}{p(w_i | \neg S)}$

4.0 Content Block

where $\propto$ denotes proportionality since the denominator $p(x)$ is omitted. This means that under the above independence assumptions, the conditional distribution over the class variable C is:

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$p(x | C_k) = \frac{(\sum_{i=1}^n x_i)!}{\prod_{i=1}^n x_i! \prod_{k=1}^n p_{ki}^{x_i}}$

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prior for a given class = $\frac{\text{no. of samples in that class}}{\text{total no. of samples}}$

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In practice, there is interest only in the numerator of that fraction, because the denominator does not depend on C and the values of the features x_i are given, so that the denominator is effectively constant. The numerator is equivalent to the joint probability model

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Parameter estimation and event models A class's prior may be calculated by assuming equiprobable classes, i.e., $p(C_k) = \frac{1}{K}$, or by calculating an estimate for the class probability from the training set:

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$$p(\text{foot size} \mid \text{male}) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(8 - \mu)^2}{2\sigma^2}\right) = 1.3112 \cdot 10^{-3}$$

10.0 Content Block

$$p(x_i, x_{i+1}, \dots, x_n, C_k) = p(x_i \mid C_k) \cdot p(x_{i+1}, \dots, x_n \mid C_k)$$

11.0 Content Block

To estimate the parameters for a feature's distribution, one must assume a distribution or generate nonparametric models for the features from the training set. The assumptions on distributions of features are called the "event model" of the naive Bayes classifier. For discrete features like the ones encountered in document classification (include spam filtering), multinomial and Bernoulli distributions are popular. These assumptions lead to two distinct models, which are often confused.

12.0 Content Block

Thus, the probability ratio $p(S \mid D) / p(\neg S \mid D)$ can be expressed in terms of a series of likelihood ratios. The actual probability $p(S \mid D)$ can be easily computed from $\log(p(S \mid D) / p(\neg S \mid D))$ based on the observation that $p(S \mid D) + p(\neg S \mid D) = 1$. Taking the logarithm of all these ratios, one obtains:

13.0 Content Block

Other heuristics "Neutral" words like "the", "a", "some", or "is" (in English), or their equivalents in other languages, can be ignored. These are also known as Stop words. More generally, some bayesian filtering filters simply ignore all the words which have a spamicity next to 0.5, as they contribute little to a good decision. The words taken into consideration are those whose spamicity is next to 0.0 (distinctive signs of legitimate messages), or next to 1.0 (distinctive signs of spam). A method can be for example to keep only those ten words, in the examined message, which have the greatest absolute value $|0.5 - p|$. Some software products take into account the fact that a given word appears several times in the examined message, others don't. Some software products use patterns (sequences of words) instead of isolated natural languages words. For example, with a "context window" of four words, they compute the spamicity of "Viagra is good for", instead of computing the spamicities of "Viagra", "is", "good", and "for". This method gives more sensitivity to context and eliminates the Bayesian noise better, at the expense of a bigger database.

14.0 Content Block

$$\log \frac{p(C_1 \mid x)}{p(C_2 \mid x)} = \log p(C_1 \mid x) - \log p(C_2 \mid x) > 0$$

15.0 Content Block

$$p(\neg S \mid D) = p(\neg S) p(D) \prod_i p(w_i \mid \neg S)$$

16.0 Content Block

Relation to logistic regression In the case of discrete inputs (indicator or frequency features for discrete events), naive Bayes classifiers form a generative-discriminative pair with multinomial logistic regression classifiers: each naive Bayes classifier can be considered a way of fitting a probability model that optimizes the joint likelihood $p(C, \mathbf{x})$, while logistic regression fits the same probability model to optimize the conditional $p(C \mid \mathbf{x})$. More formally, we have the following: